Supporting Information for : From overexploited to overshoot: how spatial protections, fishing pressure reduction and environmental conditions sparked a positive tipping point in the Adriatic common sole

Contents

Change Point Analysis

Total population

```r
## Total population
cpt.mean(solea_tot$Abun_tot,Q=1,method="BinSeg")#location 6
```

```
## Warning in BINSEG(sumstat, pen = pen.value, cost_func = costfunc, minseglen =
## minseglen, : The number of changepoints identified is Q, it is advised to
## increase Q to make sure changepoints have not been missed.
```

```
## Class 'cpt' : Changepoint Object
##        ~~   : S4 class containing 14 slots with names
##               cpts.full pen.value.full data.set cpttype method test.stat pen.type pen.value minseglen
##
## Created on  : Wed Nov 12 23:50:40 2025
##
## summary(.)  :
## ----------
## Created Using changepoint version 2.3
## Changepoint type      : Change in mean
## Method of analysis    : BinSeg
## Test Statistic  : Normal
## Type of penalty       : MBIC with value, 7.694848
## Minimum Segment Length : 1
## Maximum no. of cpts   : 1
## Changepoint Locations : 6
## Range of segmentations:
##      [,1]
## [1,]    6
##
##  For penalty values: 418722.7
```

Juveniles

```r
## Juveniles
cpt.mean(solea_tot$AbunRecruits, Q=1,method="BinSeg")#location4
```

```
## Warning in BINSEG(sumstat, pen = pen.value, cost_func = costfunc, minseglen =
## minseglen, : The number of changepoints identified is Q, it is advised to
## increase Q to make sure changepoints have not been missed.
```

```
## Class 'cpt' : Changepoint Object
##        ~~   : S4 class containing 14 slots with names
##               cpts.full pen.value.full data.set cpttype method test.stat pen.type pen.value minseglen
##
## Created on  : Wed Nov 12 23:50:40 2025
##
## summary(.)  :
```

```
## ----------
## Created Using changepoint version 2.3
## Changepoint type      : Change in mean
## Method of analysis    : BinSeg
## Test Statistic  : Normal
## Type of penalty       : MBIC with value, 7.694848
## Minimum Segment Length : 1
## Maximum no. of cpts   : 1
## Changepoint Locations : 4
## Range of segmentations:
##      [,1]
## [1,]    4
##
##  For penalty values: 53725.22
```

Adults

```
## Adults
cpt.mean(solea_tot$AbunAdults,Q=1,method="BinSeg")#location 7
```

```
## Warning in BINSEG(sumstat, pen = pen.value, cost_func = costfunc, minseglen =
## minseglen, : The number of changepoints identified is Q, it is advised to
## increase Q to make sure changepoints have not been missed.
```

```
## Class 'cpt' : Changepoint Object
##        ~~   : S4 class containing 14 slots with names
##              cpts.full pen.value.full data.set cpttype method test.stat pen.type pen.value minseglen
##
## Created on  : Wed Nov 12 23:50:40 2025
##
## summary(.)  :
## ----------
## Created Using changepoint version 2.3
## Changepoint type      : Change in mean
## Method of analysis    : BinSeg
## Test Statistic  : Normal
## Type of penalty       : MBIC with value, 7.694848
## Minimum Segment Length : 1
## Maximum no. of cpts   : 1
## Changepoint Locations : 7
## Range of segmentations:
##      [,1]
## [1,]    7
##
##  For penalty values: 201522.5
```

Bimodality

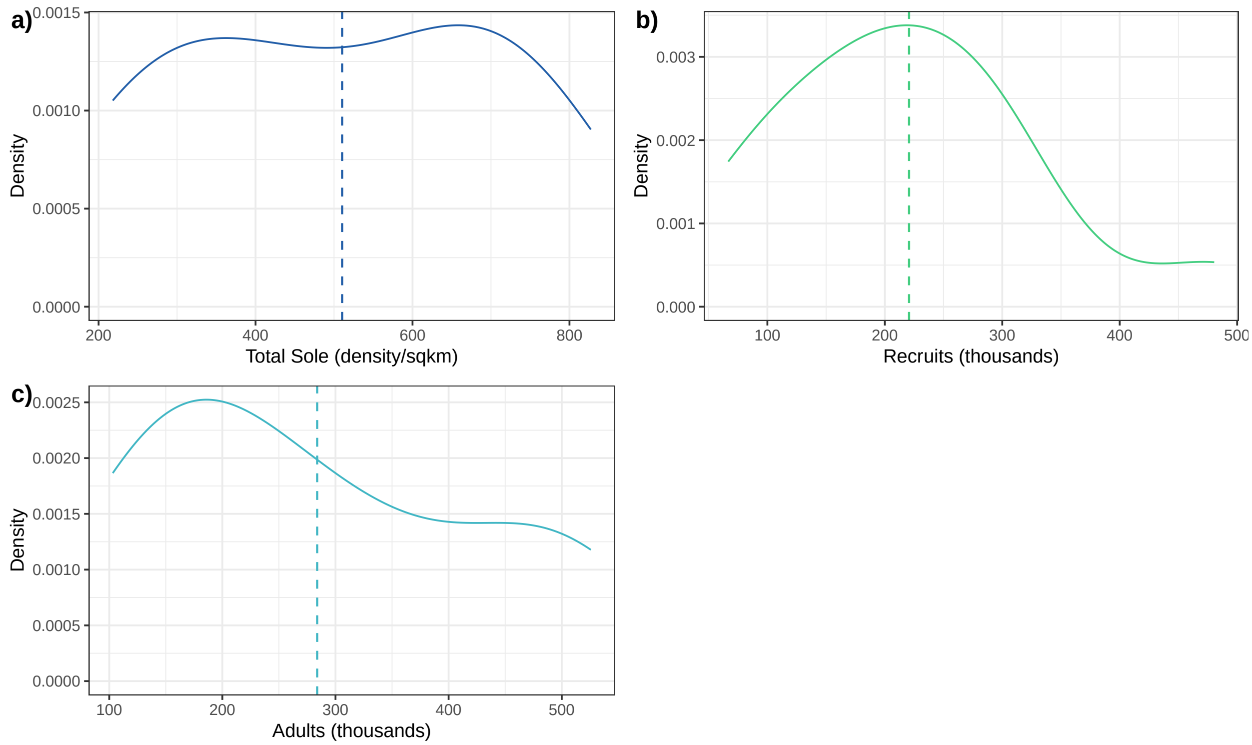

Figure A1.1: Bimodality of the response variables: a) total Common sole abundance, b) Adults abundance, c) Recruits abundance. Bimodality is given by the presence of two peaks.

Driver state plot

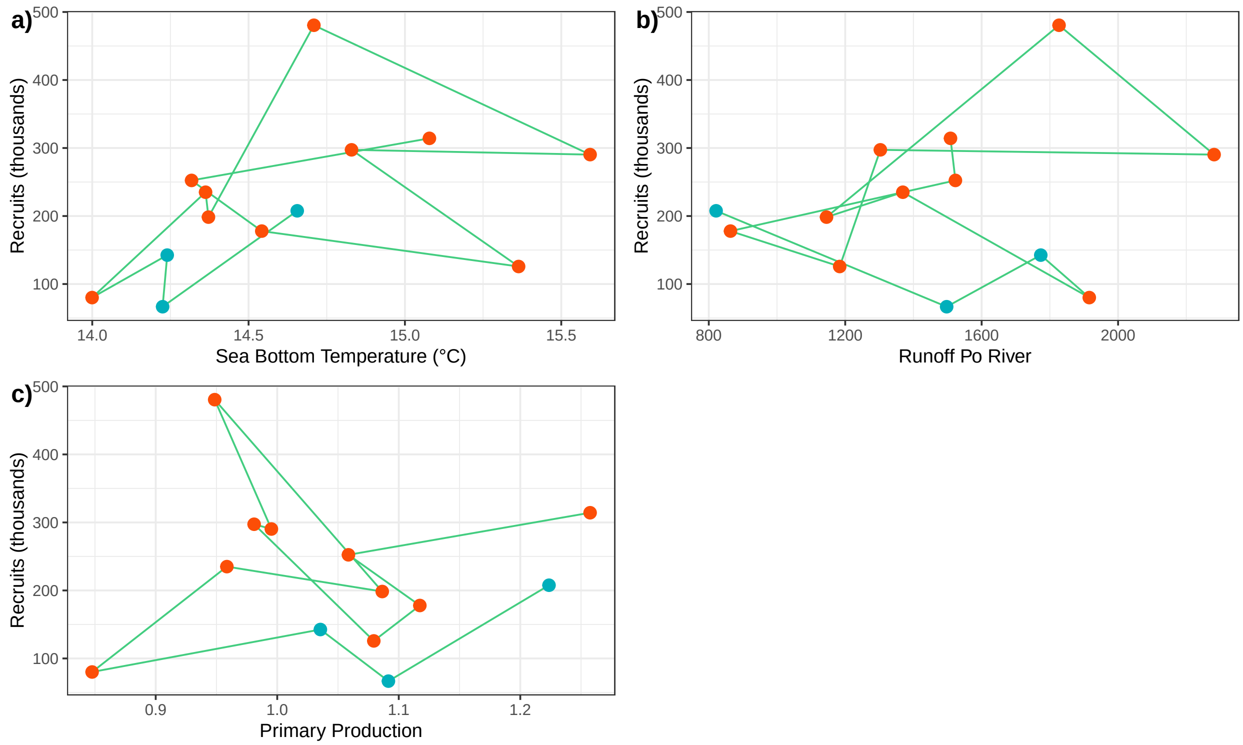

Figure A1.2: State-driver plots of recruits of Common sole in GSA 17. On the y-axis the state variable, Recruits abundance. On the x axis the a) Sea Bottom Temperature, b) Runoff of the river Po, c) Primary production. The colors refer to the periods before (blue) and after (red) the tipping point in 2012.

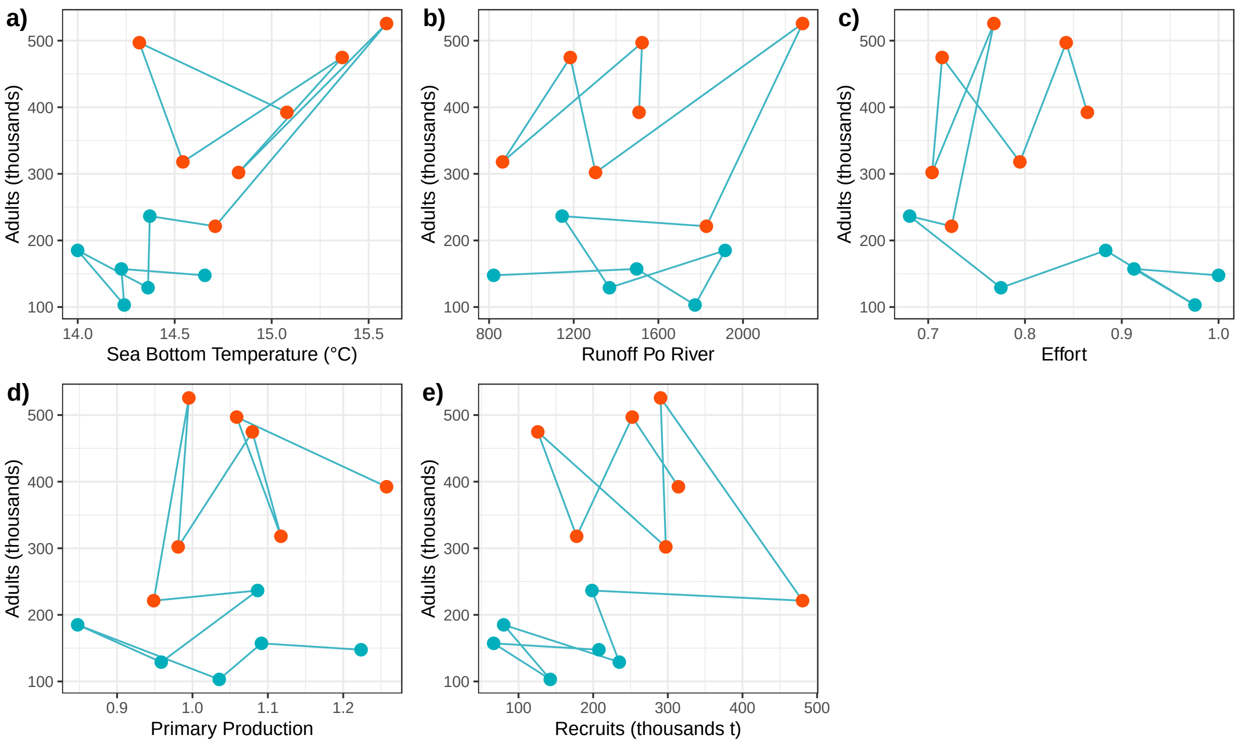

Figure A1.3: State-driver plots of total Adults of Common sole in GSA 17. On the y-axis the state variable, adults abundance. On the x axis the a) Sea Bottom Temperature, b) Runoff of the river Po, c) Primary production, d) Effort and e) recruits. The colors refer to the periods before (blue) and after (red) the tipping point in 2012.

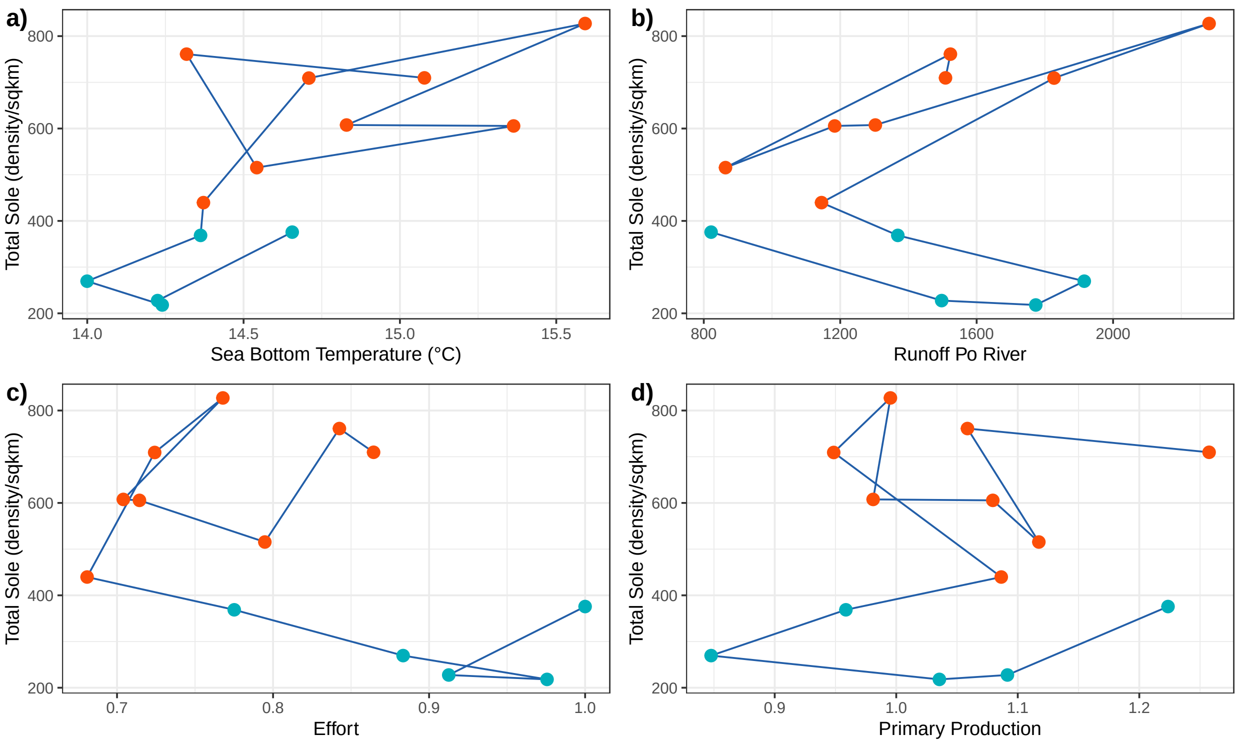

Figure A1.4: State-driver plots of total population of Common sole in GSA 17. On the y-axis the state variable, sole abundance. On the x axis the a) Sea Bottom Temperature, b) Runoff of the river Po, c) Primary production and d) Effort. The color refer to the periods before (blue) and after (red) the tipping point in 2012.

Table A1.1: Model results of linear model with Sea Bottom Temperature. The results of the model using as response variable the total population of Common Sole and as explanatory variable Sea Bottom temperature

	Estimate	St.Error	t.value	p.value	Rsq
(Intercept)	-4172.151	1363.889	-3.059	0.011	0.518
bottom_t	319.901	93.134	3.435	0.006	0.518

Table A1.2: Model results of linear model with Sea Bottom Temperature. The results of the model using as response variable the Adult population of Common Sole and as explanatory variable Sea Bottom temperature

	Estimate	St.Error	t.value	p.value	Rsq
(Intercept)	-2924.356	984.333	-2.971	0.013	0.492
bottom_t	219.176	67.216	3.261	0.008	0.492

Linear models

Here goes this trial

Table A1.3: Model results of linear model for recruits. The results of the model using as response variable the juvenile population of Common Sole and as explanatory variable Sea Bottom temperature.

	Estimate	St.Error	t.value	p.value	Rsq
(Intercept)	-1161.961	957.421	-1.214	0.250	0.16
bottom_t	94.459	65.378	1.445	0.176	0.16

Table A1.4: Model results of linear model for recruits. The results of the model using as response variable the juvenile population of Common Sole and as explanatory variable po river.

	Estimate	St.Error	t.value	p.value	Rsq
(Intercept)	137.926	119.298	1.156	0.272	0.045
runoff_ms	0.057	0.079	0.719	0.487	0.045

Table A1.5: Model results of linear model for recruits. The results of the model using as response variable the juvenile population of Common Sole and as explanatory variable primary production.

	Estimate	St.Error	t.value	p.value	Rsq
(Intercept)	239.438	318.956	0.751	0.469	0
PP	-17.829	301.524	-0.059	0.954	0

Threshold gam

Total population

```
## Formula is:
```

```
## y ~ 1 + s(x, by = I(1 * (x2 <= 0.873)), k = 4) + s(x, by = I(1 *
##     (x2 > 0.873)), k = 4)
```

```
## [[1]]
```

Adults

```
## Formula is:
```

```
## y ~ 1 + s(x, by = I(1 * (x2 <= 240.332)), k = 4) + s(x, by = I(1 *
##     (x2 > 240.332)), k = 4)
```

```
## Warning in model_resid_na[[i]][!pass_model$train_na] <- model_resid[[i]]: il
## numero di elementi da sostituire non è un multiplo della lunghezza di
## sostituzione
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

Table A1.6: Model results of linear model for recruits. The results of the model using as response variable the juvenile population of Common Sole and as explanatory variable the Adults observed in the previous year.

	Estimate	St.Error	t.value	p.value	Rsq
(Intercept)	155.129	73.199	2.119	0.060	0.072
Adult_lag	0.260	0.295	0.883	0.398	0.072

Table A1.7: Model results threshold gam for the total Common sole population. The model was built using as threshold driver effort and as the other explanatory variable the runoff of river Po.

	x
Estimate	299.371
St.Error	20.635
t.value	14.508
p.value	0.000

Table A1.8: Model results threshold gam for the total Common sole population before and after the threshold.

	D.f.	St.Error	p.value	Rsq	Dev.Expl
$<$ threshold	1.667	66.092	0.000	0.76	0.82
$>$ threshold	1.667	0.533	0.475	0.76	0.82

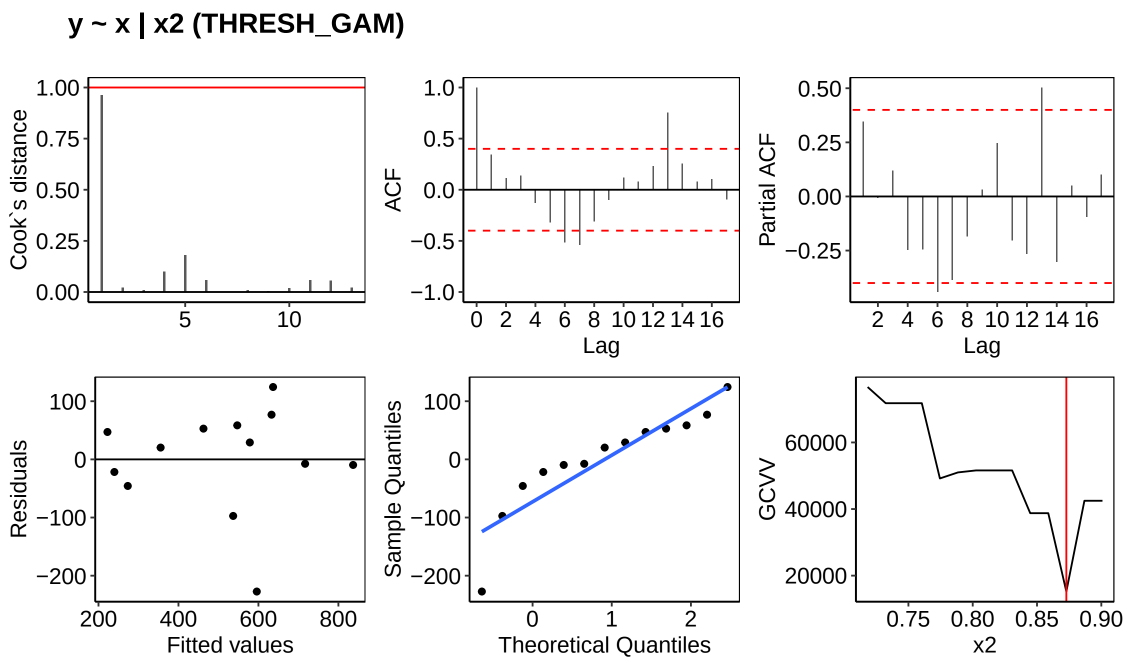

Figure A1.5: diagnostics of the threshold gam

Table A1.9: Model results threshold gam for Adults built using as threshold driver reruits and as the other explanatory variable the runoff of river Po.

	Intercept
Estimate	162.010
Sterr	41.113
tval	3.941
pval	0.006

Table A1.10: Model results threshold gam for Adults built using as threshold driver reruits and as the other explanatory variable the runoff of river Po. Model results are shown for the two parts of the model (before and after the threshold).

	D.f.	St.Error	p.value	Rsq	Dev.Expl
$<$ threshold	2.642	3.215	0.070	0.7	0.81
$>$ threshold	2.250	15.266	0.004	0.7	0.81

```
## [[1]]
```

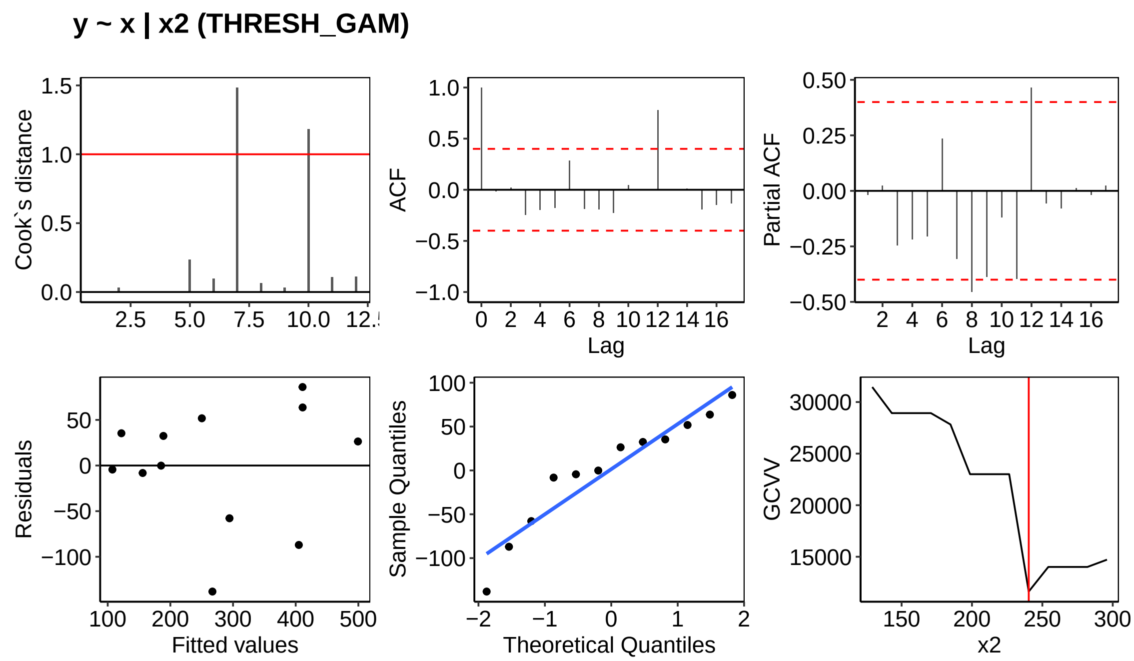

Table A1.11: Model results threshold gam for Adults built using as threshold driver reruits and as the other explanatory variable the runoff of river Po.

	Intercept
Estimate	164.698
Sterr	24.795
tval	6.642
pval	0.000

Table A1.12: Model results threshold gam for Adults built using as threshold driver reruits and as the other explanatory variable the runoff of river Po. Model results are shown for the two parts of the model (before and after the threshold).

	D.f.	St.Error	p.value	Rsq	Dev.Expl
$<$ threshold	1.667	13.432	0.005	0.3	0.475
$>$ threshold	1.667	0.061	0.937	0.3	0.475

Adults and soon

```
## Formula is:
```

```
## y ~ 1 + s(x, by = I(1 * (x2 <= 0.873)), k = 4) + s(x, by = I(1 *
##     (x2 > 0.873)), k = 4)
```

```
## Warning in model_resid_na[[i]][!pass_model$train_na] <- model_resid[[i]]: il
## numero di elementi da sostituire non è un multiplo della lunghezza di
## sostituzione
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## [[1]]
```

y ~ x | x2 (THRESH_GAM)

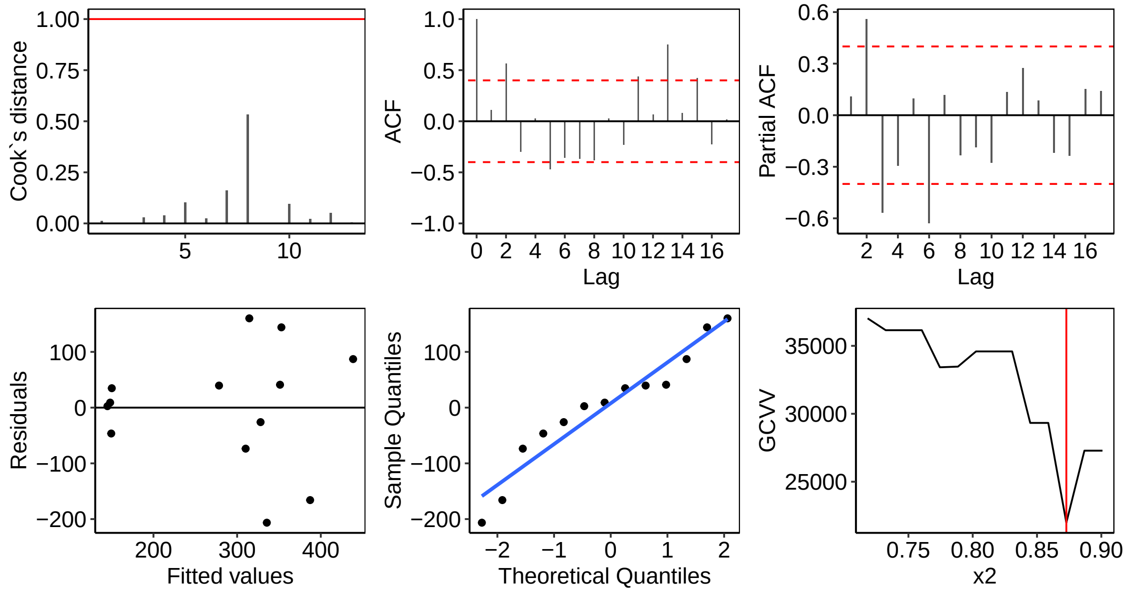

Table A1.13: Model results threshold gam for Adults built using as threshold driver reruits and as the other explanatory variable the runoff of river Po.

	Intercept
Estimate	69.634
Sterr	47.428
tval	1.468
pval	0.184

Table A1.14: Model results threshold gam for Adults built using as threshold driver reruits and as the other explanatory variable the runoff of river Po. Model results are shown for the two parts of the model (before and after the threshold).

	D.f.	St.Error	p.value	Rsq	Dev.Expl
$<$ threshold	2.581	3.039	0.144	0.711	0.824
$>$ threshold	3.018	8.297	0.010	0.711	0.824

Adults and so

```
## Formula is:
```

```
## y ~ 1 + s(x, by = I(1 * (x2 <= 14.543)), k = 4) + s(x, by = I(1 *
##     (x2 > 14.543)), k = 4)
```

```
## Warning in model_resid_na[[i]][!pass_model$train_na] <- model_resid[[i]]: il
## numero di elementi da sostituire non è un multiplo della lunghezza di
## sostituzione
```

```
## ‘geom_smooth()‘ using formula = ’y ~ x’
```

```
## [[1]]
```

y ~ x | x2 (THRESH_GAM)